

COMP1020 - Term Project Interim Report

Coffee Shop POS and Loyalty System

Team VinTony (Group 5)

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1 Project Overview

The project is a Coffee Shop Point of Sale (POS) and Loyalty System, developed as a Java desktop application. The goal is to simulate a cafe's operations—such as managing orders, customers, and inventory—by modeling real-world entities as objects and utilizing appropriate Object-Oriented Programming (OOP) principles, Design Patterns, and Data Structures.

2 Project Status & Progress

According to our proposed project timeline, we are currently in **Phase 4: Implementation**. The project is on schedule.

- **Phase 1: Planning & Proposal** (Mar 12 - Mar 26) — **[COMPLETED]**
- **Phase 2: Analysis Requirements Definition** (Mar 27 - Apr 2) — **[COMPLETED]**
- **Phase 3: Design & Prototyping** (Apr 3 - Apr 16) — **[COMPLETED]**
 - *Class diagrams, system architecture, and GUI drafts were finalized.*
- **Phase 4: Implementation** (Apr 17 - May 7) — **[IN PROGRESS]**
 - *Writing core Java code for models, patterns, and views.*
- **Phase 5: Refactoring & Feature Dev.** (May 8 - May 14) — *[UPCOMING]*
- **Phase 6: Testing & Deployment** (May 15 - May 21) — *[UPCOMING]*
- **Phase 7: Finalization & Docs** (May 22 - May 28) — *[UPCOMING]*

3 Work Completed to Date

During the design and early implementation phases, the team has successfully accomplished the following:

- **System Architecture Setup:** We have laid out the directory structure to strictly follow the MVC (**Model-View-Controller**) architecture, ensuring a clean separation between UI, business logic, and data layers.
- **OOP Class Design:** Core domain models have been established focusing on Encapsulation, Inheritance, Polymorphism, and Abstraction (e.g., **Beverage** super-class, **Customer** data protection).
- **Design Patterns Implementation:**
 - **Decorator Pattern:** Successfully implemented for handling dynamic addons. For instance, the `ToppingDecorator.java` class has been created under `src/model/addon/` to extend beverage functionalities without violating the open/closed principle.
 - **Factory Pattern:** Setup is underway in `src/factory/` to handle the dynamic creation of complex beverage objects.
- **Core Modules:** Base directories and structure for `menu`, `voucher`, `view`, and `util` have been established and are currently being populated with logic.

4 Current Focus and Next Steps

The team is currently heavily focused on executing the remainder of Phase 4. Our immediate priorities include:

- **Integrating Data Structures:** Implementing the Queue (FIFO) for order processing, HashMap for $O(1)$ lookups of customers/inventory, and Tree for hierarchical menu structuring.
- **GUI Development:** Translating our GUI prototypes into actual Java desktop interfaces in the `src/view/` package.
- **Connecting Components:** Ensuring seamless interaction between the UI layer, the backend models, and the core inventory system.

Once the core system is operational by May 7, we will transition into **Phase 5**, where we will integrate advanced features like the loyalty points and voucher system (`src/model/voucher/`), followed by rigorous testing.

5 Team Contribution & Dynamics

All 5 members of Team VinTony are actively contributing to the project. We have successfully managed the varying levels of programming experience within the team by collaborating closely. Those with prior OOP experience are helping guide the implementation of complex design patterns, ensuring that all members are learning and applying OOP fundamentals practically.

6 Conclusion

The “Coffee Shop POS and Loyalty System” project is progressing smoothly and remains on track according to our initial proposal timeline. The foundational architecture is solid, and key design patterns are actively being translated into Java code. We are confident in our ability to deliver a robust, fully functional application by the end of the term.